



ICT370 DATA ANALYTICS T126

All information in the Subject Outline is correct at the time of approval. KOI reserves the right to make changes to the Subject Outline if they become necessary. Any changes require the approval of the KOI Academic Board and will be formally advised to those students who may be affected by email and via Moodle.

Information contained within this Subject Outline applies to students enrolled in the trimester as indicated.

1. General Information

1.1 Administrative Details

Associated HE Award(s)	Duration	Level	Subject Coordinator
Bachelor of Information Technology (BIT)	1 trimester	Level 3	Dr Firoz Anwar firoz.anwar@koi.edu.au P: +61 (2) 9283 3583 L: 7-11, 11 York St. Consultation: via Moodle or by appointment.

1.2 Core / Elective

Core subject for BIT

1.3 Subject Weighting

Indicated below is the weighting of this subject and the total course points.

Subject Credit Points	Total Course Credit Points
4	BIT (96 Credit Points)

1.4 Student Workload

Indicated below is the expected student workload per week for this subject

No. Timetabled Hours/Week*	No. Personal Study Hours/Week**	Total Workload Hours/Week***
4 hours/week (2 hour Lecture + 2 hour Tutorial)	6 hours/week	10 hours/week

* Total time spent per week at lectures and tutorials

** Total time students are expected to spend per week in studying, completing assignments, etc.

*** Combination of timetable hours and personal study.

1.5 Mode of Delivery Classes will be face-to-face or hybrid. Certain classes will be online (e.g., special arrangements).

1.6 Pre-requisites BUS105 Business Statistics

1.7 General Study and Resource Requirements

- Dedicated computer laboratories are available for student use. Normally, tutorial classes are conducted in the computer laboratories.
- Students are expected to attend classes with the requisite textbook and must read specific chapters prior to each tutorial. This will allow them to actively take part in discussions. Students should have elementary skills in both word processing and electronic spreadsheet software, such as Office 365 or MS Word.



- Computers and WIFI facilities are extensively available for student use throughout KOI. Students are encouraged to make use of the campus Library for reference materials.
- Students will require access to the internet and email. Where students use their own computers, they should have internet access. KOI will provide access to required software.

Resource requirements specific to this subject: MS Imagine, MS Azure, Office 365, R-3.4.4 for windows.

1.8 Academic Advising

Academic advising is available to students throughout teaching periods including the exam weeks. As well as requesting help during scheduled class times, students have the following options:

- Consultation times: A list of consultation hours is provided on the homepage of Moodle where appointments can be booked.
- Subject coordinator: Subject coordinators are available for contact via email. The email address of the subject coordinator is provided at the top of this subject outline.
- Academic staff: Lecturers and Tutors provide their contact details in Moodle for the specific subject. In most cases, this will be via email. Some subjects may also provide a discussion forum where questions can be raised.
- Head of Program: The Head of Program is available to all students in the program if they need advice about their studies and KOI procedures.
- Vice President (Academic): The Vice President (Academic) will assist students to resolve complex issues (but may refer students to the relevant lecturers for detailed academic advice).

2 Academic Details

2.1 Overview of the Subject

Data is everywhere. The amount of digital data that exists is growing at a rapid rate, doubling every two years, and is thus drastically changing the way we live. Massive amounts of data and information are collected every second by the internet, social media, sensors and networks all over the world. Business Intelligence (BI) frameworks must handle this data. Business intelligence requires data science that deals with data cleansing, preparing, integrating and analysing data sets in order to draw conclusions about the information they contain. This subject focuses on the analysis and reporting of data and its applications to business intelligence. The subject helps you to combine technical and statistical skills, analytical thinking, and business acumen.

2.2 Graduate Attributes for Undergraduate Courses

Graduates of Bachelor courses from King's Own Institute (KOI) will achieve the graduate attributes expected under the Australian Qualifications Framework (2nd edition, January 2013). Graduates at this level will be able to apply a broad and coherent body of knowledge from their major area of study in a range of contexts for professional practice or scholarship and as a pathway for further learning.

King's Own Institute's generic graduate attributes for a bachelor's level degree are summarised below:

	KOI Bachelor Degree Graduate Attributes	Detailed Description
	Knowledge	Current, comprehensive, and coherent and connected knowledge
	Critical Thinking	Critical thinking and creative skills to analyse and synthesise information and evaluate new problems

	Communication	Communication skills for effective reading, writing, listening and presenting in varied modes and contexts and for transferring knowledge and skills to a variety of audiences
	Information Literacy	Information and technological skills for accessing, evaluating, managing and using information professionally
	Problem Solving Skills	Skills to apply logical and creative thinking to solve problems and evaluate solutions
	Ethical and Cultural Sensitivity	Appreciation of ethical principles, cultural sensitivity and social responsibility, both personally and professionally
	Teamwork	Leadership and teamwork skills to collaborate, inspire colleagues and manage responsibly with positive results
	Professional Skills	Professional skills to exercise judgement in planning, problem solving and decision making

Across the course, these skills are developed progressively at three levels:

- **Level 1 Foundation** – Students learn the basic skills, theories and techniques of the subject and apply them in basic, standalone contexts
- **Level 2 Intermediate** – Students further develop the skills, theories and techniques of the subject and apply them in more complex contexts, and begin to integrate this application with other subjects.
- **Level 3 Advanced** – Students demonstrate an ability to plan, research and apply the skills, theories and techniques of the subject in complex situations, integrating the subject content with a range of other subject disciplines within the context of the course.

2.3 Subject Learning Outcomes

This is a Level 3 subject.

On successful completion of this subject, students should be able to:

Subject Learning Outcomes	Contribution to Graduate Attributes
a) Justify the business demand for data collection, decision support and data analytics	   
b) Analyse and evaluate data using different data analytics layers and techniques	   
c) Evaluate modelling techniques in the context of analytics and Big Data	  
d) Develop and select data hypotheses and methods and evaluate analytics models. Furthermore, apply these models to business intelligence scenarios.	   
e) Demonstrate in-depth analytical, data mining and machine learning domain knowledge that is based on well-founded principles	    
f) Develop, evaluate and justify appropriate visualisation approaches from a range of applicable options for stakeholders.	    



2.4 Subject Content and Structure

Below are details of the subject content and how it is structured, including specific topics covered in lectures and tutorials. Reading refers to the text unless otherwise indicated.

Weekly Planner:

Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 1 02 March	An Overview of Business Intelligence and Analytics	Ch 1	Tutorial 1 Discussion questions and exercises to explore resources in Business context. Formative weekly tutorial
Week 2 09 March	Analytics Ecosystem and Big Data introduction	Ch 1	Tutorial 2 Discussion questions and exercises to review Case Studies and Data sets and prepare summaries Biweekly activity 0%
Week 3 16 March	Nature of Data and Statistical Modeling	Ch 2	Tutorial 3 Discussion questions and exercises to do basic statistical analysis and modelling
Week 4 23 March	Business Reporting and Data Visualization	Ch 2	Tutorial 4 Discussion questions and exercises to build basic reports and visualizations using tools Biweekly activity 5%
Week 5 30 March	Business Intelligence and Data Warehousing	Ch 3	Tutorial 5 Discussion questions and exercises to understand data warehousing latest trends and developments
Week 6 07 April (Tue)	Data Mining Concepts and Applications	Ch 4	Tutorial 6 Discussion questions and exercises to practice some applications of data mining Biweekly activity 5%
Week 7 13 April	Text Analytics and Text Mining	Ch 5	Tutorial 7 Discussion questions and exercises related to using Text Analytics tools and establishing findings Quiz-20%
Week 8 20 April	Web Mining and Social Analytics	Ch 5	Tutorial 8 Discussion questions and exercises to related to using Web Analytics tools and establishing findings Biweekly activity 5%



Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 9 28 April (Tue)	Manage legal, ethical, privacy and security issues related to the use of data	Ch 8	Tutorial 9 Discussion questions and exercises related to legality, privacy, ethics and security
Week 10 04 May	Prescriptive Analytics: Optimization and Simulation	Ch. 6	Tutorial 10 Discussion questions and exercises for decision modelling Biweekly activity 5%
Week 11 11 May	Big Data Concepts and Tools	Ch. 7	Tutorial 11 Discussion questions and exercises related to researching the Big Data concepts and Tools Assessment 3 Due Presentation 25%
Week 12 18 May	Future Trends in Analytics	Ch. 8	Tutorial 12 Discussion questions and exercises related to future trends in Analytics Assessment 4 Due Report 35%
Week 13 25 May	Study review week and Final Exam Week		
Week 14 01 June	Examinations Continuing students - enrolments for T226 open		Please see exam timetable for exam date, time and location
Week 15 09 June (Tue)	Student Vacation begins New students - enrolments for T226 open		
Week 16 15 June	• Results Released • Review of Grade Day for T126 – see Sections 2.6 and 3.2 below for relevant information. • Certification of Grades NOTE: More information about the dates will be provided at a later date through Moodle/KOI email.		
T226 29 June 2026			
Week 1 29 June	Week 1 of classes for T226		

2.5 Public Holiday Amendments

Please note: KOI is closed on all scheduled NSW Public Holidays.

T126 has Five (5) public holidays that occur during this trimester. Classes scheduled for these public holidays (Calendar Class Date) will be rescheduled as per the table below.

This applies to ALL subjects taught in T126.



Please see the table below and adjust your class timing as required. Please make sure you have arrangements in place to attend the rescheduled classes if applicable to your T126 enrolment.

Classes will be conducted at the same time and in the same location as your normally scheduled class except these classes will be held on the date shown below.

Calendar Class Date	Rescheduled Class Date
03 April 2026 (Friday) 04 April 2026 (Saturday) 06 April 2026 (Monday) 25 April 2026 (Saturday) 27 April 2026 (Monday)	25 May 2026 (Monday) 26 May 2026 (Tuesday) 27 May 2026 (Wednesday)

2.6 Review of Grade, Deferred Exams & Supplementary Exams/Assessments

Review of Grade Day:

There may be instances when you believe that your final grade in a subject does not accurately reflect your performance against the marking criteria. Section 8 of the *Assessment and Assessment Appeals Policy* (www.koi.edu.au) describes the grounds on which you may apply for a Review of Grade.

If you have a concern about your marks and you are unable to resolve it with the Academic staff concerned, then you can apply for a formal Review of Grade as explained in section 3.2(e) Appeals Process below. Please note the time limits for requesting a review. Please ensure you read the Review of Grade information before submitting an application.

Review of Grade Day:

Final exam scripts will not normally be returned to students. Students can obtain feedback on their exam performance and their results for the whole subject at the Review of Grade Day. KOI will hold the Review of Grade Day for all subjects studied in T126. **The ROG day will be announced at a later date and the students will be notified through Moodle/KOI email.**

Only final exams and whole subject results will be discussed as all other assessments should have been reviewed during the trimester. Further information about Review of Grade Day will be available through Moodle.

If you fail one or more subjects and you wish to consider applying for a Review of Grade you are **STRONGLY ADVISED** to attend the Review of Grade Day. You will have the chance to discuss your final exam and subject result with your lecturer, and will be advised if you have valid reasons for applying for a Review of Grade (see Section 3.2 below and the *Assessment and Assessment Appeals Policy*).

A formal request for a review of grade may not be considered unless you first contact the subject coordinator to discuss the result.

Deferred Exams:

If you wish to apply for a deferred exam because you are unable to attend the scheduled exam, you should submit the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days of the assessment due date*.

If you miss your final exam there is no guarantee you will be offered a deferred exam.

You must apply within the stated timeframe and satisfy the conditions for approval to be offered a deferred exam (see Section 8.1 of the *Assessment and Assessment Appeals Policy and the Application for Assignment Extension or Deferred Exam Forms*). In assessing your request for a deferred exam, KOI will take into account the information you provide, the severity of the event or circumstance, your performance on other items of assessment in the subject, class attendance and your history of previous applications for special consideration.

Deferred Quizzes will be held before the end of week 9. Deferred final exams will be held before the next



trimester. You will not normally be granted a deferred exam on the grounds that you mistook the time, date or place of an examination, or that you have made arrangements to be elsewhere at that time; for example, have booked plane tickets.

If you are offered a deferred exam, but do not attend *you will be awarded 0 marks for the exam*. This may mean it becomes difficult for you to pass the subject. If you apply for a deferred exam within the required time frame and satisfy the conditions you will be advised by email (to your KOI student email address) of the time and date for the deferred exam. Please ensure that you are available to take the exam at this time.

Marks awarded for the deferred exam will be the marks awarded for that item of assessment towards your final mark in the subject.

Supplementary Assessments (Exams and Assessments):

A supplementary assessment may be offered to students to provide a final opportunity to demonstrate successful achievement of the learning outcomes of a subject. *Supplementary assessments are only offered at the discretion of the Board of Examiners*. In considering whether or not to offer a supplementary assessment, KOI will take into account your performance on all the major assessment items in the subject, your attendance, participation and your history of any previous special considerations.

If you are offered a supplementary assessment, you will be advised by email to your *KOI student email address* of the time and due date for the supplementary assessment – supplementary exams will normally be held at the same time as deferred final exams during week 1 or week 2 of the next trimester.

You must pass the supplementary assessment to pass the subject. The maximum grade you can achieve in a subject based on a supplementary assessment is a PASS grade.

If you:

- are offered a supplementary assessment, but fail it;
- are offered a supplementary exam, but do not attend; or
- are offered a supplementary assessment but do not submit by the due date; you will receive a FAIL grade for the subject.

Students are also eligible for a supplementary assessment for their final subject in a course where they fail the subject but have successfully completed all other subjects in the course. You must have completed all major assessment tasks for the subject and obtained a passing mark on at least one of the major assessment tasks to be eligible for a supplementary assessment.

If you believe you meet the criteria for a supplementary assessment for the final subject in your course, but have not received an offer, complete the *Complaint, Grievance, Appeal Form* and send your form to reception@koi.edu.au. The deadline for applying for supplementary assessment is the Friday of the first week of classes in the next trimester.

2.7 Teaching Methods/Strategies

Briefly described below are the teaching methods/strategies used in this subject:

- *Lectures* (2 hours/week) are conducted in seminar style and address the subject content, provide motivation and context and draw on the students' experience and preparatory reading.
- *Tutorials* (2 hours/week) include class discussion of case studies and research papers, practice sets and problem-solving and syndicate work on group projects. Tutorials often include group exercises and so contribute to the development of teamwork skills and cultural understanding. Tutorial participation is an essential component of the subject and contributes to the development of many of the graduate attributes (see section 2.2 above). Tutorial participation contributes towards the assessment in many subjects (see details in Section 3.1 for this subject). Supplementary tutorial material such as case studies, recommended readings, review questions etc. will be made available each week in Moodle.
- *Online* teaching resources include class materials, readings, model answers to assignments and exercises and discussion boards. All online materials for this subject as provided by KOI will be found in the Moodle



page for this subject. Students should access Moodle regularly as material may be updated at any time during the trimester

- **Other contact** - academic staff may also contact students either via Moodle messaging, or via email to the email address provided to KOI on enrolment.

2.8 Student Assessment

Assessment is designed to encourage effective student learning and enable students to develop and demonstrate the skills and knowledge identified in the subject learning outcomes. Assessment tasks during the first half of the study period are usually intended to maximise the developmental function of assessment (formative assessment). These assessment tasks include weekly tutorial exercises (as indicated in the weekly planner) and low stakes graded assessment (as shown in the graded assessment table). The major assessment tasks where students demonstrate their knowledge and skills (summative assessment) generally occur later in the study period. These are the major graded assessment items shown in the graded assessment table.

Final grades are awarded by the Board of Examiners in accordance with KOI's Assessment and Assessment Appeals Policy. The definitions and guidelines for the awarding of final grades within the BIT degree are:

- HD High distinction (85-100%) an outstanding level of achievement in relation to the assessment process.
- DI Distinction (75-84%) a high level of achievement in relation to the assessment process.
- CR Credit (65-74%) a better than satisfactory level of achievement in relation to the assessment process.
- P Pass (50-64%) a satisfactory level of achievement in relation to the assessment process.
- F Fail (0-49%) an unsatisfactory level of achievement in relation to the assessment process.

Provided below is a schedule of formal assessment tasks and major examinations for the subject.

Assessment Type	When assessed	Weighting	Learning Outcomes Assessed
Assignment 1 – Biweekly (4,6,8 and 10) Individual Progress Reports and Reflection - Individual Assignment	Week 4- 5% Week 6- 5% Week 8 - 5% Week 10 - 5%	20%	a, b, c, d, e, f
Assignment 2: Quiz	Week 7	20%	a, b
Assignment 3: Business Case Study Group Presentation	Week 11	25%	b, c, d, e, f
Assignment 4: Case Study Group Report (2,000 words)	Week 12	35%	b, c, d, e, f

Requirements to Pass the Subject:

To gain a pass or better in this subject, students must gain a *minimum of 50%* of the total available subject marks.

2.9 Prescribed and Recommended Readings

Provided below, in formal reference format, is a list of the prescribed and recommended readings.

Prescribed Text:

Sharda, R., Delen, D., Turban, E. 2017, *Business intelligence, analytics, and data science: a managerial perspective*, 4th Global edn, Pearson, Harlow, UK. Available from: ProQuest Ebook Central. ISBN: 9781292220543.

Recommended Texts:

Sarker, I.H., 2021, Data science and analytics: An overview from data-driven smart computing, decision-making and applications perspective, *SN Computer Science*, vol. 2, no. 5, p. 377. doi:10.1007/s42979-021-00765-8.

Teng, Y., Zhang, J., and Sun, T., 2022, Data-driven decision-making model based on artificial intelligence in higher education system of colleges and universities, *Expert Systems*. Vol. 40, no. 4, doi:10.1111/exsy.12820.

Tayeb, K.M., 2023, Decision-making and data science: How large businesses can use analytics to shape decisions, *Business & IT*, vol. 13, no. 2, pp. 55–64. doi:10.14311/bit.2023.02.06.

Sarker, I.H., Kayes, A.S.M., Badsha, S., Alqahtani, H., Watters, P., and Ng, A., 2020, Cybersecurity data science: An overview from machine learning perspective, *Journal of Big Data*, vol. 7, no. 1, pp. 1–29. <https://doi.org/10.1186/s40537-020-00318-5>

Dasgupta, N., 2018, *Practical big data analytics: Hands-on techniques to implement enterprise analytics and machine learning using Hadoop, Spark, NoSQL and R*, Packt Publishing, Birmingham. ISBN: 9781783554393.

Sulova, S., 2021, Text mining approach for identifying research trends, *Proceedings of the International Conference on Computer Systems and Technologies 21*. doi:10.1145/3472410.3472433.

Yan, H., Ma, M., Wu, Y., Fan, H., and Dong, C., 2022, Overview and analysis of the text mining applications in the construction industry, *Heliyon*, vol. 8, no. 12, e12088. doi:10.1016/j.heliyon.2022.e12088.

The state of AI in 2023: Generative AI's breakout year August 1, 2023 | Survey <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-in-2023-generative-ais-breakout-year>

Recommended Journal Articles:

Calof, J., Richards, G., and Santilli, P., 2024, Integration of business intelligence with corporate strategic management, *Journal of Intelligence Studies in Business*, vol. 7, no. 3, pp. 62–73. doi:10.37380/jisib.v7i3.280.

Charkaoui, A., and Jabraoui, S., 2024, 20 years of scientific study on business intelligence and decision-making performance: A bibliometric analysis, *Journal of Information Systems Engineering and Business Intelligence*, vol. 10, no. 3, pp. 408–421. doi:10.20473/jisebi.10.3.408-421.

Kasneci, E., Seßler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., and Kasneci, G., 2023, ChatGPT for good? On opportunities and challenges of large language models for education, *Learning and Individual Differences*, vol. 103, p. 102274. <https://doi.org/10.1016/j.lindif.2023.102274>.

Szukits, Á., 2022, The illusion of data-driven decision making - The mediating effect of digital orientation and controllers' added value in explaining organizational implications of advanced analytics, *Journal of Management Control*, vol. 33, no. 3, pp. 403–446. <https://doi.org/10.1007/s00187-022-00343-w>.

Ozaydin, B., Zengul, F., Oner, N., and Feldman, S.S., 2020, Healthcare Research and Analytics Data Infrastructure Solution: A Data Warehouse for Health Services Research, *Journal of Medical Internet Research*, vol. 22, no. 6, pp. e18579–e18579. <https://doi.org/10.2196/18579>.

Sarker, I.H., 2021, Data science and analytics: An overview from data-driven smart computing, decision-making and applications perspective, *SN Computer Science*, vol. 2, no. 5, p. 377. doi:10.1007/s42979-021-00765-8.

Pohl, M., Staegemann, D.G., and Turowski, K., 2022, The performance benefit of data analytics applications, *Procedia Computer Science*, vol. 201, pp. 679–683. <https://doi.org/10.1016/j.procs.2022.03.090>.

Online Resources:

Analytics Magazine(Digital Edition) <http://analytics-magazine.org/digital-editions/>
Several Reading at <https://learning-academics.teradata.com/> requires student registration using KOI email
<https://www.linkedin.com/learning/the-essential-elements-of-predictive-analytics-and-data-mining/>
<https://www.ibm.com/products/cognos-analytics>
<https://www.tableau.com/learn/articles/what-is-predictive-analytics>
<https://learn.microsoft.com/en-us/power-bi/connect-data/service-tutorial-build-machine-learning-model>

Journals:

- Analytics Magazine (Digital Edition) <http://analytics-magazine.org/digital-editions/>
- Journal of Data Science
- Journal of Emerging Technologies in Web Intelligence

Conference/ Journal Articles:

Razmerita, L. 2024, Human-AI collaboration: A student-centered perspective of generative AI use in higher education, *European Conference on e-Learning*, vol. 23, no. 1, pp. 320–329. doi:10.34190/ecel.23.1.3008.

Lin, C.Y., Lobo Marques, J.A. & Chan, L.K. 2024, Artificial intelligence and deep learning in stock prediction: A bibliometric review, *European Conference on Management Leadership and Governance*, vol. 20, no. 1, pp. 297–306. doi:10.34190/ecmlg.20.1.3003.

Basha, M.J., Murthy, T.S., Valarmathy, A.S., Abbas, A.R., Gavhar, D., Rajavarman, R. & Parkunam, N. 2023, Privacy-preserving data mining and analytics in big data, *E3S Web of Conferences*, vol. 399, p. 04033. <https://doi.org/10.1051/e3sconf/202339904033>

Students are encouraged to read peer reviewed journal articles and conference papers. Google Scholar provides a simple way to broadly search for scholarly literature. From one place, you can search across many disciplines and sources: articles, theses, books, abstracts and court opinions, from academic publishers, professional societies, online repositories, universities and other web sites.

Useful Websites:

The following websites are useful sources covering a range of information useful for this subject. However, most are not considered to be sources of Academic Peer Reviewed theory and research. If your assessments require **academic peer reviewed journal articles** as sources, you need to access such sources using the Library database, Ebscohost, or Google Scholar. Please ask in the Library if you are unsure how to access Ebscohost. Instructions can also be found in Moodle.

- TeraData University Network <https://learning-academics.teradata.com/>
- Analytics, Big Data, and IT Research <https://tdwi.org/research/list/research-and-resources.aspx>
- Data Science Central <https://www.datasciencecentral.com/>
- Open Source Intelligence Australia <https://osint.net.au>
- <https://www.datacenterdynamics.com/en/>
- <https://www.datasciencecentral.com/>

3. Assessment Details

3.1 Details of Each Assessment Item

The assessments for this subject are described below. Supplementary assessment information and assistance can be found in Moodle.

KOI expects students to submit their own original work in both assignments and exams, or the original work of their group in the case of group assignments.

Marking guides for assessments follow the assessment descriptions. Students should compare final drafts of their assessment against the marking guide before submission.



Assessment 1

Assessment type: Individual Progress Report and Reflection (300-500 words)

Purpose: This biweekly report is where the student can record their progress with the concepts and tools being applied in the course and reflect upon their observations and responses to situations, which can then be used to explore and improve further analysis. The purpose of the assessment is to test your understanding of analytical principles being applied in different categories and confirms your usage of tools. The aim of a reflective log is to give you an opportunity to keep a record of the work you undertake, note any existing skills you develop, and learn to identify areas in which you would like to improve. This assessment contributes to learning outcomes a, b, c, d, e, f.

Value: 20% **Due Date:** Biweekly submission
Week 4 (5%) + Week 6 (5%) + Week 8 (5%) + Week 10 (5%) = Total (20%)

Assessment topic: Progress Report and Reflection on two weeks in class activities

Task Details: The requirements for each biweekly assessment will be provided on Moodle within below categories.

Categories for the biweekly report

- data set selection
- data preparation
- applying tools for statistical analysis
- applying tools for visualisations
- applying analytics for novel findings

Submission requirements details: Students will be expected to prepare the reports including screen shots and submit within the discussion groups on Moodle that will be due biweekly.

Marking Rubric for Assessment 1 (5% Biweekly)

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Evidence of Learning 1 mark	The report shows little, or no evidence of learning tied to concepts and tools covered.	The report documents some, but not sufficient, learning tied to concepts and tools covered.	The report partially documents learning tied to concepts and tools covered.	The portfolio provides evidence of learning tied to concepts and tools covered.	The portfolio provides clear evidence of learning tied to sound understanding of concepts and tools covered.
Justification for work done 1 mark	Justification is Incorrect, Irrelevant or missing.	Answer is unreasonable with major flaws and unclear explanations.	Appropriate justification provided for some of the completed tasks.	Clear and detailed justification provided for most of the completed tasks.	Accurate and Thorough explanation provided for all the completed tasks



Content Reflection 2 marks	Reflection lacks critical thinking. Superficial connections are made with key concepts and tools covered.	Reflection demonstrates limited critical thinking in applying, analysing, and/or evaluating key concepts and tools covered. Minimal connections made	Reflection demonstrates some degree of critical thinking in applying, analysing, and/or evaluating key concepts and tools covered. Some connections made through explanations and/or inferences	Reflection demonstrates a high degree of critical thinking in applying, analysing, and evaluating key concepts and tools covered. Good connections made through related explanations and/or examples.	Reflection demonstrates an exceptional degree of critical thinking in applying, analysing, and evaluating key concepts and tools covered. Insightful and relevant connections made through contextual explanations, inferences, and examples.
Progress 1 marks	Conveys inadequate evidence of reflection on own work in response to the self-assessment questions posed. Personal growth and awareness are not evident and/or demonstrates a neutral experience with negligible personal impact. Lacks sufficient inferences, examples, personal insights and challenges, and/or future implications are overlooked.	Conveys limited evidence of reflection on own work in response to the self-assessment questions posed. Demonstrates less than adequate personal growth and awareness through few or simplistic inferences made, examples, insights, and/or challenges that are not well developed. Minimal thought of the future implications of current experience.	Conveys evidence of reflection on own work with a personal response to the self-assessment questions posed. Demonstrates satisfactory personal growth and awareness through some inferences made, examples, insights, and challenges. Some thought of the future implications of current experience.	Conveys strong evidence of reflection on own work with a personal response to the self-assessment questions posed. Demonstrates significant personal growth and awareness of deeper meaning through inferences made, examples, well developed insights, and substantial depth in perceptions and challenges. Synthesizes current experience into future implications.	Conveys exceptionally strong evidence of reflection on own work with a personal response to the self-assessment questions posed. Demonstrates significant personal growth and awareness of deeper meaning through inferences made, examples, well developed insights, and substantial depth in perceptions and challenges. Synthesizes current experience into future implications.

Assessment 2

Assessment type: In-Class Quiz

Purpose: Covers topics of Weeks 1 to 6. This assessment contributes to learning outcomes a, and b.

Value: 20%

Due Date: Week 7

Submission requirements details: Quiz will be conducted **in class** and completed online via Moodle.

- Attend their enrolled class **in person** to do the quiz, unless their enrollment allows online attendance.
- Students with online enrollment must ensure their computer has a working **safe exam browser** (SEB) and working camera.

Assessment topic:

Week 1 – Week 6 materials.

Task Details: The quiz will consist of a series of short answer questions relating to subject content weeks 1 – 6 inclusive.

Marking Rubric: Assessment 2

Value:20%



	Unsatisfactory	Satisfactory	Good	Very good	Exceptional
Grade	Fail	Pass	Credit	Distinction	High distinction
Marks	0-49%	50-64%	65-74%	75-84%	>84%

Assessment 3 and Assessment 4:

Assessment type: Group project - Presentation (Assessment 3) and Report (Assessment 4)

Purpose: The purpose of this assessment is that students will learn about different topics relevant to data analytics by carrying out research and also by listening to presentations made by their peers. This assessment contributes to learning outcomes b, c, d, e, f

Value: Total 60% (Presentation 25%, Report 35%)

Due Date: Weeks 11 – 12

Assessment topic: Students to select a data source and suggest a topic of analysis for that data source. Tutors to approve the topic before students proceed with further data preparation and analytics

Students will work in groups (minimum 3 and maximum 4 students in each group).

The first step will be to identify a data set from one of the publicly available data sets and present the summary of some of the target models in Descriptive and Predictive Analytic layers to the tutor for approval. Once approved by their tutor, they will further define the research questions and prepare the data using consolidation and reduction if needed.

Next, they will select one of the analytical tools (e.g. Excel, Tableau, Rapid Miner) and apply analytical methods for generating novel findings and draw insights from this data set. These outcomes need to be presented using visualisation models and also need to be explained in a detailed report.

Students will present their findings as a group during tutorial sessions in week 11 for a duration of 10-15 mins per group. Tutors will provide feedback on their findings and students will then need to update their findings to reflect this feedback in their group report.

Submission of a group report will be due in Week 12. This will be a 2,000 words report excluding references and executive summary.

Along with the basic components, the report should also include

- Justification for dataset selection and any ethics, privacy and security related concerns
- List of research questions (e.g. what novel findings are you targeting?, what are the relationship with some of the features/ variables etc.)
- Data readiness (e.g. what steps did you take to reduce/clean/integrate data, how did you handle missing values, did you perform any normalisation etc.)
- Data Analysis Findings:
 - Summary of findings
 - Literature review of relevant Data Analytical techniques, data mining approaches, machine learning models, application of such models in real world and case studies
 - Descriptive analysis results along with appropriate visualisations (**at least 5 different visualisations to support your findings**)
 - Predictive analysis results along with supporting visualisations (**at least 2 predictive analysis**)
 - Compare your findings with other published researches and/or case studies
- Response to Feedback and make appropriate adjustments to incorporate the suggested changes given during the presentation demonstration
- Ethics, security and privacy discussion
- List of suggestions to provide improvements in data set or findings



- Your report must have at least 8 - 10 references in the correct referencing format.
- Submission requirements details: Report should be uploaded via Turnitin on Moodle in Week 12. The presentation conducted by the group should be uploaded on Moodle in week 11.

Task details: Marking Rubrics Assessment 3: Presentation (25%)

Criteria	Fail (0 – 49%)	Pass (50 - 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Visual appeal (group) 5 mark	There are many errors in spelling, grammar, and punctuation, the slides are difficult to read and contain too much text, poor choice of fonts and colors, no or little visual appeal	There are errors in spelling, grammar, and punctuation, too much text on many slides, minimal effort made to make slides appealing	There may be some errors in spelling, grammar and punctuation, too much text on two or more slides, significant visual appeal	There are no errors in spelling, grammar, and punctuation, information is clear and concise on each slide, visually appealing and engaging	No errors, created engaging and professional looking presentation
Content (group) 8 mark	The presentation provides a brief look at the topic but many questions are left unanswered, majority of information is irrelevant and significant points left out	The presentation is informative but several elements are unanswered, much of the information irrelevant, coverage of some of major points	The presentation is a good summary of the topic, most important information has been analysed, little irrelevant information	The presentation is a concise summary of the topic with all questions answered, comprehensive and complete coverage of information	Exceptionally good summary of the topic and provides extensive supportive elements to aid the ease of understanding of the audience
Preparedness/ participation/ group dynamics (group) 5 marks	Unbalanced presentation or tension resulting from over-helping. multiple group members not participating, evident lack of preparation/rehearsal, dependence on slides	Significant controlling by some members with one minimally contributing, primarily prepared but with some dependence on just reading off slides	Slight predominance of One presenter, Members help each other, very well prepared	All presenters know the information, participated equally and help each other as needed, extremely well prepared and rehearsed	Exceptionally good group dynamics, all members coherently presented in a professional manner
Presentation skills (individual) 7 marks	Minimal eye contact focusing on small part of audience, the audience is not engaged, spoke too quickly or quietly making it difficult to understand, poor body language	Focuses on only part of the audience, sporadic eye contact and the audience is distracted, speaker could be heard by only half of the audience, body language is distracting	Speaks to majority of the audience, steady eye contact, the audience is engaged by the presentation, speaks at a suitable volume, minor problems with body language eg. fidgeting	Regular/constant eye contact, the audience is engaged, and presenter held the audience's attention, appropriate speaking volume and good body language	Professional presentation skills, excellent audience engagement with clear understanding of the concepts

Marking Rubrics Assessment 4: Report (35%)

Report	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Structure and layout 3 marks	Very difficult to read, structure not clear.	Some difficulty in reading, structure lacking in some parts	Well written, structure not totally clear	Well written and structured.	Very clearly written and well-structured with subheadings



Introduction 3 marks	Introduces the data set and organisation of the report, but omits a general background of the topic and/or the overall "plan" of the report	Satisfactorily introduces the data set and organisation of the report, gives a general background. Indicates the overall "plan" of the report	Introduces the data set and organisation of the report in an engaging manner to some extent which arouses the reader's interest	Introduces the data set and organisation of the report in an engaging manner which arouses the reader's interest, gives some general background and indicates the overall "plan" of the report	Introduces the data set and organisation in an extremely engaging manner which arouses the reader's interest, gives a detailed general background and indicates the overall "plan" of the report
Data Analysis 20 marks	Inadequate discussion, little/no demonstrated understanding or analysis of most issues and/or some irrelevant information	Most parts are adequately discussed, displays some understanding and analysis of issues, some justification for choices provided	Consistently reasonable discussion of data analysis techniques and findings, displays a reasonable evaluation of relevant issues and justification for choices provided	Consistently detailed discussion of the data analysis techniques and findings, displays sound analysis and evaluation of relevant issues and justification for choices provided	Data analysis techniques and findings are discussed in depth, displays a deep analysis of relevant issues and shows creativity in the justification for choices provided
Ethics, privacy, security 3	No analysis provided for ethical, privacy and security concerns	Ethical, privacy and security issues are cited but relevancy is not clear.	Correctly recognizes and applies ethical, privacy and security concept(s)	Analysis establishes and maintains focus on ethical, privacy and security considerations	Correctly applied ethical constructs and relevant privacy issues, also evaluated data security
Response to feedback 3 marks	Feedback is not considered at all	Some of the feedback responded to, requires more details and justification	Most of the feedback responded to, requires more details and justification	All of the feedback responded, could include more details and justification	All of the feedback responded to with comprehensive details and justification
Conclusion 3 marks	Poor/no summary of the main points and irrelevant or no recommendations	Satisfactory summary of the main points, a final comment on the subject but introduced new material, some recommendations provided but incomplete	A summary of the main points, a reasonable final comment on the subject, based on the information provided, good recommendations provided	A good summary of the main points, a good final comment on the subject, based on the information provided, recommendations cover most issues	An interesting, well written summary of the main points, an excellent final comment on the subject, based on the information provided, recommendations cover all issues

3.2 General information about assessment

a) Late Penalties and Extensions

An important part of business life and key to achieving KOI's graduate outcome of Professional Skills is the ability to manage workloads and meet deadlines. Completing assessment tasks on time is a good way to master these habits.

Students who miss final exams without a valid and accepted reason may not be granted a deferred exam and will be awarded 0 marks for the assessment item. Assessment items which are missed or submitted after the due date/time will attract a penalty unless there is a compelling reason (see below). These penalties are designed to encourage students to develop good time management practices, and to create equity for all students.

Any penalties applied will only be up to the maximum marks available for the specific piece of assessment attracting the penalty.

Late penalties, granting of extensions and deferred exams are based on the following:

In Class Tests and Quizzes

- Generally, extensions are not permitted. A make-up test may only be permitted under very special circumstances where acceptable supporting evidence of illness, hardship or unavoidable problems preventing completion of the assessment is provided (see section (b) below). The procedures and timing to



apply for a make-up test (only if available) are as shown in the section *Applying for an Extension* (see below).

- Missing a class test will result in 0 marks for that assessment item unless the above applies.

Written Assessments and Video Assessments

- There is a late penalty of *5% of the total available marks* per calendar day unless an extension is approved (see *Applying for an Extension* section below).

Presentations

- Generally, extensions are not permitted. Missing a presentation will result in 0 marks for that assessment item. The rules for make-up presentations are the same as for missing in-class tests (described above). For group presentations, if serious circumstances prevent some members of the group from participating, the members of the group who are present should make their contributions as agreed. If a make-up presentation is approved, the other members of the group will be able to make their individual presentation later and will be marked according to the marking rubric. A video presentation may be used to facilitate the process.

Final Exams

If students are unable to attend final exams due to illness, hardship or some other unavoidable problem (acceptable to KOI), they must:

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days after the exam date*.
- Provide acceptable documentary evidence (see section (b) below).
- Agree to attend the deferred exam as set by KOI if a deferred exam is approved.

Deferred exam

- There will only be one deferred exam offered.
- Marks obtained for the deferred exam will be the marks awarded for that assessment item.
- If you miss the deferred exam you will be awarded 0 marks for the assessment item. This may mean you are unable to pass the subject.

b) Applying for an Extension

If students are unable to submit or attend an assessment when due, they must

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days of the assessment due date*.
- Provide acceptable documentary evidence in the form of a medical certificate, police report or some other appropriate evidence of illness or hardship, or a technician's report on problems with computer or communications technology, or a signed and witnessed statutory declaration explaining the circumstances.
- Students and lecturers / tutors will be advised of the outcome of the extension request as soon as practicable.

Please remember there is no guarantee of an extension being granted, and poor organisation is not a satisfactory reason to be granted an extension.

c) Referencing and Plagiarism

Please remember that all sources used in assessment tasks must be suitably referenced.

Failure to acknowledge sources is plagiarism, and as such is a very serious academic issue. Students plagiarising run the risk of severe penalties ranging from a reduction in marks through to 0 marks for a first offence for a single assessment task, to exclusion from KOI in the most serious repeat cases. Exclusion has serious visa implications. The easiest way to avoid plagiarising is to reference all sources.

Harvard referencing is the required method – in-text referencing using Author's Surname (family name) and year



of publication. A Referencing Guide, "Harvard Referencing", and a Referencing Tutorial can be found on the right-hand menu strip in Moodle on all subject pages.

An effective way to reference correctly is to use *Microsoft Word's* referencing function (please note that other versions and programs are likely to be different). To use the referencing function, click on the References Tab in the menu ribbon – students should choose *Harvard*.

Authorship is also an issue under plagiarism – KOI expects students to submit their own original work in both assessment and exams, or the original work of their group in the case of a group project. All students agree to a statement of authorship when submitting assessments online via Moodle, stating that the work submitted is their own original work.

The following are examples of academic misconduct and can attract severe penalties:

- Handing in work created by someone else (without acknowledgement), whether copied from another student, written by someone else, or from any published or electronic source, is fraud, and falls under the general Plagiarism guidelines.
- Copying / cheating in tests and exams is academic misconduct. Such incidents will be treated just as seriously as other forms of plagiarism.
- Students who willingly allow another student to copy their work in any assessment may be considered to assisting in copying/cheating, and similar penalties may be applied.

Where a subject coordinator considers that a student might have engaged in academic misconduct, KOI may require the student to undertake an additional oral exam as a part of the assessment for the subject, as a way of testing the student's understanding of their work.

Further information can be found on the KOI website.

d) Reasonable Adjustment

The Commonwealth Disability Discrimination Act (1992) makes it unlawful to treat people with a disability less fairly than people without a disability. In the context of this subject, the principle of Reasonable Adjustment is applied to ensure that participants with a disability have equitable access to all aspects of the learning for the subject. For assessment, this means that barriers to their demonstrating competence are removed wherever it is reasonably practical to do so.

Examples of reasonable adjustment in assessment may include:

- provision of an oral assessment, rather than a written assessment
- provision of extra time
- use of adaptive technology.

The focus of the adjusted assessment should be on enabling the student to demonstrate achievement of the learning outcomes for the subject, rather than on the method of assessment.

e) Appeals Process

Full details of the KOI *Assessment and Assessment Appeals Policy* may be obtained in hard copy from the Library, and on the KOI website www.koi.edu.au under *Policies and Forms*.

Assessments :

Where students are not satisfied with the results of an assessment, they have the right to appeal. The process is as follows:

- Discuss the assessment with their tutor or lecturer – students should identify where they feel more marks should have been awarded – students should provide valid reasons based on the marking guide provided for the assessment. Reasons such as "*I worked really hard*" are not considered valid.
- If still not satisfied, students should complete an Application for Review of Assessment Marks form, clearly



explaining the reasons for seeking a review. This form is available from the KOI website under *Policies and Forms* and is also available at KOI Reception (Kent St, Market St and O'Connell St). The completed Application for Review of Assessment Marks form should be submitted as explained on the form with supporting evidence attached to academic@koi.edu.au.

- The form must be submitted within *ten (10) working days* of the return of the marked assessment, or within *five (5) working days* after the return of the assessment if the assessment is returned after the end of the trimester.

Review of Grade – whole of subject and final exams:

Where students are not satisfied with the results of the whole subject or with their final exam results, they have the right to request a Review of Grade – see the *Assessment and Assessment Appeals Policy* for more information.

An *Application for Review of Grade/Assessment Form* (available from the KOI Website under *Policies and Forms* and from KOI Reception at Kent St, Market St and O'Connell St) should be completed clearly explaining the grounds for the application. The completed application should be submitted as explained on the form, with supporting evidence attached to academic@koi.edu.au.